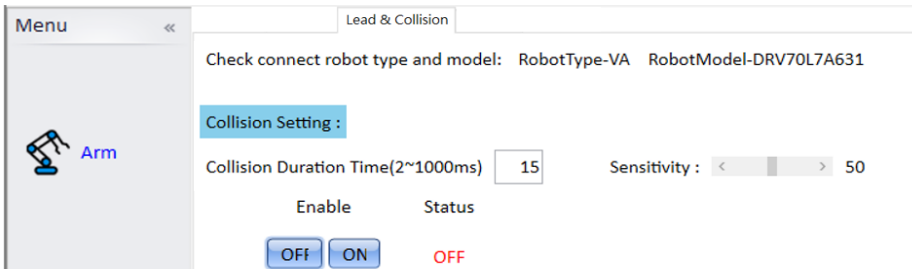


8.3 Lead & Collision

In the **Lead & Collision** tab, you can set the parameters for collision protection to reduce damage to the robot when there is a collision.

8.3.1 Collision Setting

When detecting a collision, the collision protection brakes the motor and reverses the robot moving direction to prevent the robot from continuously moving to the target position, or it may cause more serious damage.



| Item                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|----|----------------------------------|----|-----------------------------|-----|----------------------|-----|---------------------------------|-----|--------------------|-----|--------------------|-----|--------------------|-----|----------------------------------|-----|------------------|
| Collision Duration Time | Set the delay time of triggering the collision protection after a collision occurs. Setting range: 2 to 1000; unit: ms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| Sensitivity             | Set the detection sensitivity in the range of 1 to 100. The greater the setting value, the higher the sensitivity; in that case, a minor bump or slight contact triggers the collision protection. Conversely, only a major collision triggers the collision protection. Adjust the sensitivity as required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| Enable                  | <b>OFF</b> : switch off the collision protection.<br><b>ON</b> : switch on the collision protection.<br>When the Mass setting in the <b>Payload</b> tab does not match the actual weight, it affects the collision detection. Thus, make sure you have properly set the Mass before switching on the collision protection.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| Status                  | Shows whether the collision protection is ON. You can also go to the <b>Status</b> subpage and click <b>Update</b> to check the status for "Collision Detect". <div><div>Status</div><div><div>Update</div><table><tr><th>Status</th><th>Content</th></tr><tr><td>ON</td><td>Gravity Compensation Deformation</td></tr><tr><td>ON</td><td>Gravity Compensation Torque</td></tr><tr><td>OFF</td><td>Thermal Compensation</td></tr><tr><td>OFF</td><td>FigureAngle Compensation(SCARA)</td></tr><tr><td>OFF</td><td>System Reservation</td></tr><tr><td>OFF</td><td>System Reservation</td></tr><tr><td>OFF</td><td>System Reservation</td></tr><tr><td>OFF</td><td>FigureLength Compensation(SCARA)</td></tr><tr><td>OFF</td><td>Collision Detect</td></tr></table></div></div> | Status | Content | ON | Gravity Compensation Deformation | ON | Gravity Compensation Torque | OFF | Thermal Compensation | OFF | FigureAngle Compensation(SCARA) | OFF | System Reservation | OFF | System Reservation | OFF | System Reservation | OFF | FigureLength Compensation(SCARA) | OFF | Collision Detect |
| Status                  | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| ON                      | Gravity Compensation Deformation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| ON                      | Gravity Compensation Torque                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| OFF                     | Thermal Compensation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| OFF                     | FigureAngle Compensation(SCARA)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| OFF                     | System Reservation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| OFF                     | System Reservation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| OFF                     | System Reservation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| OFF                     | FigureLength Compensation(SCARA)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |
| OFF                     | Collision Detect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |         |    |                                  |    |                             |     |                      |     |                                 |     |                    |     |                    |     |                    |     |                                  |     |                  |